

Advanced Real Estate Price Prediction

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Problem Statement

Property-specific attributes, broader economic indicators, and neighborhood-level factors influence real estate prices in the United States. Traditional machine learning models often fail to capture the complex interactions among these variables. This causes inaccurate valuations that can misguide buyers, sellers, and policymakers, and potentially destabilize the local housing market, even the local economy.

This project aims to build an advanced machine learning model that integrates property, neighborhood, and macroeconomic data to deliver more accurate and transparent valuations. Improved predictions can enhance pricing efficiency, reduce risks of artificial inflation or deflation, and help build fairer markets. The model can also support economists in tracking housing trends and help policymakers design effective housing and economic policies. In the long term, it offers more chances for people to own a house and helps the local economy grow healthily.

Data Sources

Our project integrates multiple publicly available datasets to develop a predictive and interpretable model for housing value trends across the United States. Each dataset contributes a unique dimension - economic, social to explain and forecast the Zillow Home Value Index (ZHVI).

1. [Zillow Dataset](#) :
Structure: Monthly home value estimates at varying geographic levels (ZIP code, county, city).
Relevance: Captures housing market trends and affordability. We will focus on ZIP code-level data to analyze the relationship between local real estate values, income levels, and crime rates.
2. [FBI Crime Dataset](#):
Structure: Counts of crimes and arrests over the years, disaggregated by offense type.
Relevance: Provides insight into community safety and social conditions.
3. [Yelp Open Dataset](#) :
Structure: JSON files containing business data along with customer reviews.
Relevance: Provides insight into local business activity and community engagement. By linking businesses to neighborhoods, we can examine how the presence of local businesses may affect housing demand, neighborhood appeal, and ultimately home values.
4. [Median Household Income](#) :
Structure: Annual estimates of median household income by geography (county, ZIP code).
Relevance: Median household income helps us understand the economic conditions of a neighborhood, which can influence housing demand and home values over time.

All the datasets we are using are publicly available and do not include personal information. The main challenges will be cleaning and combining the data, since they come in different formats, cover different areas, and span different time periods. We will also need to process large files like Yelp's JSON data. Doing this will let us create a single dataset to study how income, crime, and local businesses relate to housing trends.

Methodology

We will implement a multi-stage pipeline to integrate four heterogeneous datasets. The Zillow Home Value Index (ZHVI) serves as our target variable, providing monthly housing prices at the zip code level from 2015 to 2024.

- 1) **Regression:** We begin with Linear Regression to establish an interpretable baseline, followed by Lasso/Ridge Regression to handle multicollinearity between correlated features (e.g., crime rates and income levels).
- 2) **Trees:** Random Forest and Gradient Boosting (XGBoost) will capture non-linear relationships and feature interactions that linear models miss. Random Forest's ensemble of decision trees naturally handles mixed data types (continuous ZHVI prices, zip codes, count-based crime data)
- 3) **Deep Learning for Sequential Dependencies:** LSTM (Long Short-Term Memory) networks will process the sequential monthly data, learning long-term dependencies that traditional time-series methods might miss. The LSTM architecture will use (Will change according to performance): Input sequences will be 5-year rolling windows of prices, crime, and business metrics with Hidden layers of 2 LSTM layers (128, 64 units) with dropout for regularization, and the Output will be next month's price prediction.
- 4) **Attention-Based Architecture:** We implement a Transformer architecture with attention mechanisms as our most advanced model. This addresses limitations of RNNs/LSTMs by: Parallel processing of entire time sequences, Self-attention to identify which historical time points most influence current prices, Cross-attention between different data modalities (correlating crime spikes with price drops).

Validation Strategy: Test/Train splits + cross validation in training set
Dataset Split: 70% for training, 15% for validation, and 15% for testing.

Project Management

- 9/23-9/30: Collect datasets, preprocess. check-in: 9/30, datasets are cleaned for feature analysis.
- 10/1-10/7: Data analysis, identifying correlations among features, etc. Prepare for model design. Status check-in: 10/7, all the information needed for the model designed needs to be prepared.
- 10/8-10/17: Design the model, start training. Status check-in: 10/17. Finish model design.
- 10/18-10/21: Prepare for a status report 1. Review the completed process and make improvements. Status check-in: 10/20. The report needs to be finished and present on 10/21.
- 10/22-11/15: Training the model, evaluating and improving the model. Status check-in: 11/15. Generally finish the model training progress.
- 11/16-11/18: Prepare for a status report 2. Review the completed process and make improvements. Status check-in: 11/17. The report needs to be finished and present on 11/18.
- 11/19-11/25: Finish model training. Status check-in: 11/25. Get the final model after the training.
- 11/26-12/4: Write draft for final report and PPT. Status check-in: 12/4. Complete the drafts.
- 12/5-12/9: Improve the report and PPT, prepare for the final presentation. Status check-in: on 12/8. Get the final version of the report and PPT. Ready for the presentation 12/9.

Risk 1: The model may be more complex than expected, requiring additional resources and time for training. We will track training time with predefined limits and reserve extra time in the schedule. ASU Sol is prepared to support training in emergency situations. To mitigate this risk, we will balance model performance with training efficiency during the design stage.

Risk 2: Since each stage depends on the previous one, delays can cascade through the project. We will monitor task completion numbers daily. Extra time is prepared for each stage to handle emergency situations. Tasks may be transferred between members to make sure they can be finished on time. To mitigate this risk, team members will update their progress in the group chat every day, and raise questions promptly so the team can resolve issues quickly and minimize delays.